

Mastering Azure Repos

MARS | 1 Day



Master Git version control with Azure Repos, from everyday commits to advanced history rewriting and repository administration. This one-day course builds your fluency across Git concepts, IDE integration, branching and merging, pull requests, and the power features that round out Azure Repos. You'll work in teams, in a shared project on a common codebase, putting each capability to use the way a real software team would.

Who should take this course?

This course is for all software developers using or considering Azure Repos for Git version control, whether new to Git or sharpening advanced skills. It's equally valuable for tech leads, build engineers, and others responsible for repository configuration and branching strategy. Some version control experience is recommended. Experience with Git, Visual Studio or Visual Studio Code, and C# is helpful but not required.

Course content

This course progressively builds your fluency with Azure Repos, from Git fundamentals and IDE integration to branching, merging, pull requests, and advanced repository administration. You'll work hands-on in teams on a common codebase, the way a real software team would.

1. INTRODUCTION TO AZURE REPOS

- Azure DevOps and Azure Repos overview
- Public vs. private repos
- Creating, configuring, and securing a project
- Creating and configuring a repository
- Hands-on lab

2. GIT CONCEPTS

- DVCS concepts and Git overview
- Using Git from the command line
- Creating/cloning a repository
- Git commit and history
- Basic Git workflows
- Hands-on lab

3. IDE INTEGRATION

- Visual Studio and VS Code integration
- Connecting to a repository
- Basic Git workflows revisited
- Viewing history and resolving conflicts
- Associating work items with commits
- Ignoring files with .gitignore
- Hands-on lab

4. WORKING WITH AZURE REPOS

- Inspecting a repository: status, tags, and blame
- Discarding and removing changes
- Checking out a previous version
- Stashing changes
- Reverting and resetting
- Rewriting history: amend, rebase, and reflog
- Branching and merging
- Squash merging
- Pull requests and code reviews
- Branch policies
- Hands-on lab

5. MASTERING AZURE REPOS

- Git hooks and service hooks
- Azure Repos extensions and Code Search
- Forking and the forking workflow
- GitHub integration
- Git workflows (Feature Branch, GitFlow, GitHub Flow, trunk-based)
- Scalar and partial clone for large repos
- Hands-on lab

Course Designer

This course was designed by Richard Hundhausen, Microsoft's first Visual Studio ALM/DevOps MVP, a Professional Scrum Trainer, and co-creator of the Nexus framework, with decades helping teams build and ship software. To see other Azure DevOps and GitHub courses, visit www.accentient.com.